

Inductive and Deductive Based Learning

- There are basically two methods for knowledge extractions firstly domain experts and then with machine learning.
- For a very large amount of data, the domain expert are not very useful and reliable.
- So, we move towards the machine learning approach for this work.
- Also called as Deterministic Supervised Learning.
- In this first input x , (the verified value) given to a function (f), and the output is $f(x)$.
- Then we can give diff sets of inputs (raw I/P) to the same functⁿ f , and verify the output $f(x)$.
- By using O/P we generate (learn) the rules.

Inductive Learning also known as. Discovery Learning, is a process where the learner discovers rules by observing examples.

- We can often work out rules for ourselves by observing examples. If there is a pattern; then record it.
- We then apply the rule in diff situations to see if it works.
- With Inductive lang learning, tasks are designed

specifically to guide the learner. and assist them in discovering a rule.

System tries to make a "general Rule" from a set of observed Instances.

eg. Mango $\rightarrow f(\text{mango}) \rightarrow \text{sweet } (e_1)$

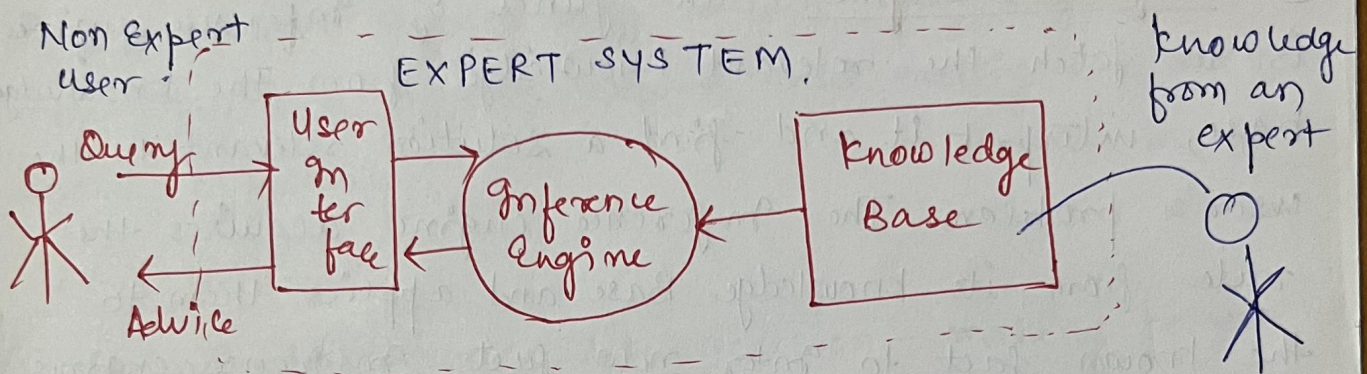
Banana $\rightarrow f(\text{banana}) \rightarrow \text{sweet } (e_2)$

Fruits $\rightarrow f(\text{fruits}) \rightarrow \text{sweet (general rule)}$

Game Play
NLP
system

Unit-3

Game Playing :- scope of AI Games, Theorem proving, NLP, vision and speech processing, Robotics, expert system, AI-Techniques - search knowledge, Abstraction.



An expert System is AI software that uses knowledge stored in a knowledge base to solve problems that would usually require a human expert thus preserving a human expert's knowl. in its knowledge Base. They can advise users as well to as provide explanations to them about how they reached a particular conclusion or advice.

Knowledge Engineering :- is the term to define the process of building an Expert System and its practitioners are called knowledge Engineer.

① Knowledge Base! - The knowledge base represents facts and rules. It consists of knowledge in a particular domain as well as rules to solve a problem, procedures and intrinsic data relevant to the domain.

② Inference Engine! - The function of inference engine is to fetch the relevant knowledge from the knowledge base, interpret it and find a solution relevant to the user's problem. The inference engine acquires the rule from its knowledge base and applies them to the known fact to enter new facts. Inference engines can also include an explanation and debugging abilities.

③ Knowledge Acquisition and Learning Module! - The function of this component is to allow the expert system to acquire more and more knowledge from various sources and store it in the knowledge base.

④ User Interface! - This module makes it possible for a non-expert user to interact with the expert system and find a solution to the problem.

⑤ Explanation Module! - This module helps the expert system to give the user an explanation about how the expert system reached a particular conclusion.

Characteristics of an Expert System:-

- ① Human experts are perishable, but an expert system is permanent.
- ② It helps to distribute the expertise of a human.
- ③ One expert system may contain knowledge from more than one human experts thus making the system more efficient.

Limitations

- ① Do not have human-like decision-making power.
- ② Cannot possess human capabilities.
- ③ Cannot produce correct result from less amount of knowledge.
- ④ Requires excess training.

Adv

- ① Low Accessibility cost
- ② Fast Response
- ③ Low Error Rate
- ④ Capable of explaining how they reached a solution.

Disadv

- ① has no emotions
- ② Common sense Issue
- ③ Developed for a specific Domain.
- ④ Needs to be updated Manually

Applications of AI Unit-4

① Medical Sciences

AI in medical, uses machine learning models to search medical data and uncover insights to help improve health outcomes and patient experiences.

② AI is quickly becoming the integral part of modern healthcare.

③ AI are being used to support medical professionals in clinical settings and in ongoing research.

④ The most common roles for AI in medical sciences are clinical decision support and imaging analysis.

⑤ Clinical decision support tools help providers make decisions about treatments, medications, mental health and other patient needs by providing them with quick access to information that is relevant to the patient.

⑥ In Medical, AI tools are being used to analyze CT scans, X-rays, MRIs and other images that a human radiologist might miss.

→ ⑦ AI has the potential to provide customized real-time recommendations to patients.

→ ⑧ AI-powered virtual assistant that could answer questions based on the patient's medical story, preferences and personal needs.

Few exhibit...
to learn a sun...
to

② Environmental Science

- ① It is possible to use satellites to detect phytoplankton presence in the ocean due to concentration in the surface. However even using satellites is not an easy task as aerosol concentration, suspended sediments give rise to complication.
- ② To overcome this problem, ML Algorithms such as ANN & Support Vector Machine are used to automatically classify & detect phytoplankton in the ocean.
- ③ It is also possible to use remote sensing to classify and detect land cover applying the same algorithm to identify and classify different types of vegetation including forests, dead trees etc. These are some of samples of AI in Environmental Science.

Explanation Based Learning

Explanation based learning is a type of machine learning that uses a very strong or even flawless, domain theory to generalise or create ideas from training data.

→ EBL is a moral strategy for improving supervised learning by utilising Available Domain knowledge. As a result, the rate of learning, the certainty of learning, precision of the taught concept, can be enhanced.

→ It is a method for an Intelligent System to learn by seeing instances. The capacity of EBL Systems to generate justified generalisations from single training instances distinguishes them.

→ EBL can learn from just one training event.

Rather than taking multiple examples, EBL emphasises learning a single, specific model.

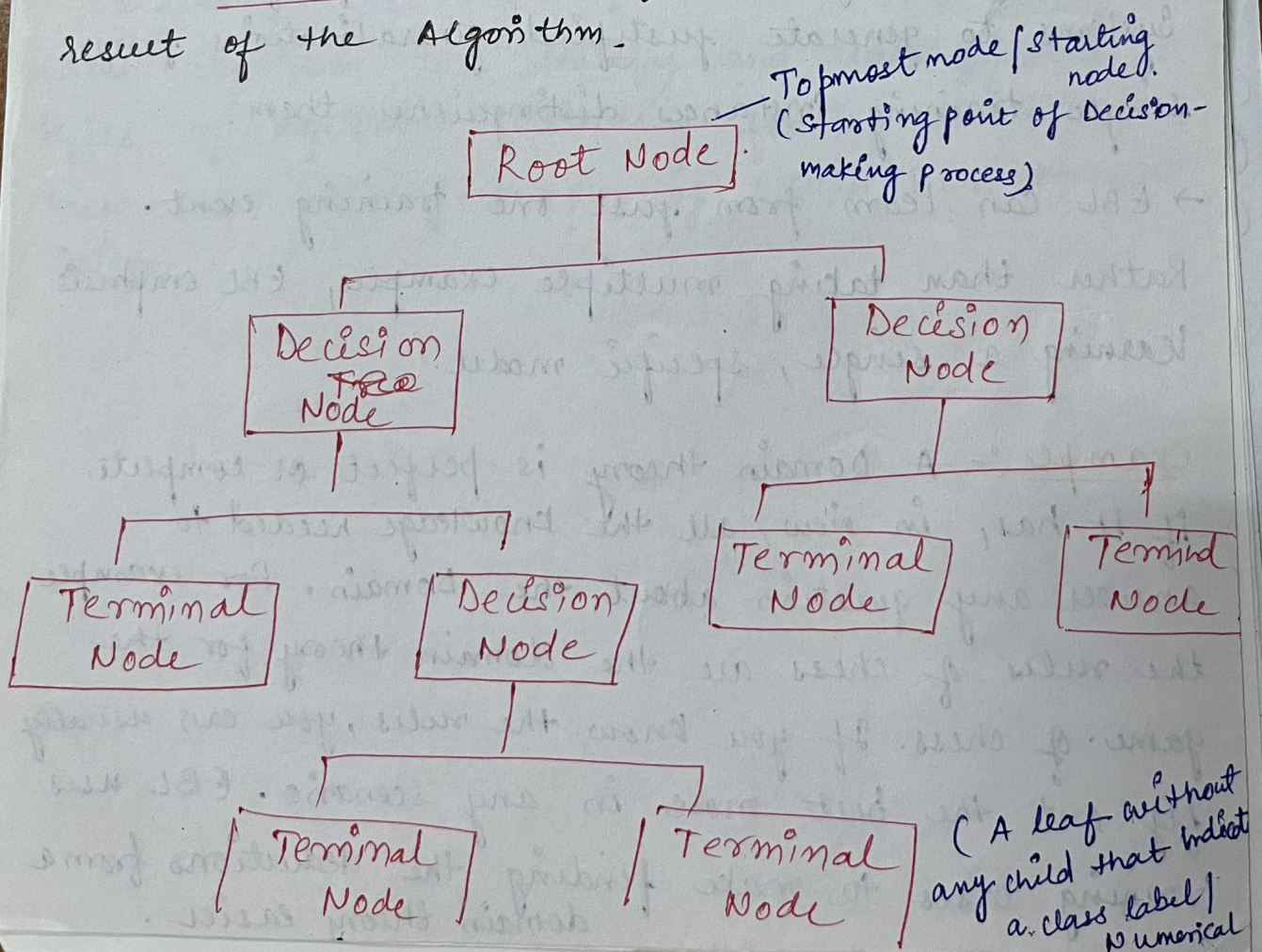
→ example:- A Domain theory is perfect or complete if it has, in view, all the knowledge needed to answer any question about the Domain. For example the rules of chess are the domain theory for the game of chess. If you know the rules, you can usually fig out the best move in any scenario. EBL uses training cases to make finding the deductions from a domain theory easier.

Decision Tree

A decision tree is one of the most powerful tools of supervised learning algo. used for both classification and regression.

→ It builds a flowchart like tree structure where each internal node denotes a test on ~~the~~ an attribute, each branch represent an outcome of the test, each leaf node (terminal node) holds a class label.

A decision tree is flowchart-like tree structure where each internal node denotes the feature, branches denote the rules and leaf nodes denotes the result of the Algorithm.



Understanding PEAS in AI

PEAS stands for Performance Measure, Environment, Actuator, and Sensor.

① Performance Measure:- is the unit to define the success of an agent. Performance varies with agents based on their different Percepts.

② Environment:- is the surrounding of an agent at every instant. It keeps changing with time if the agent is set in motion. There are 5 major types of environment.

- ① Fully observable & Partially observable
- ② Episodic & Sequential
- ③ Static & Dynamic
- ④ Discrete & Continuous
- ⑤ Deterministic & Stochastic

③ Actuator:- An actuator is a part of agent that delivers the output of action to the environment.

④ Sensors:- Sensors are the receptive part of an agent that takes in the input for the agent.

Few expert system examples

- ① MYCIN :- One of the earliest expert system based on Backward chaining. It can identify various bacteria that can cause severe infections and also recommend drugs based on person's weight.
- ② DENDRAL :- It was an AI based expert system used for chemical analysis. It used a substance's Spectrographic data to predict its molecular structure.
- ③ RI/XCON :- It could select specific S/W to generate a computer system wished by the user.
- ④ PXDES :- It could easily determine the type and degree of lung cancer in a patient based on the data.
- ⑤ Cadet :- It is a clinical support system that could identify cancer in its early stages in patients.
- ⑥ DXexplain :- It was a clinical support that could suggest a variety of disease based on the finding of the doctor.

Computational Learning Theory

Computational learning theory is a subfield of AI, that deals with the design and analysis of Machine learning algorithms. The goal of Comp. learning theory is to understand the Computational properties of these algorithm including their ability to learn from data and generalise to new data.

Goals of Computational Theory

It is concerned with how computers can learn from data. The main goals of Computational learning theory are to develop algorithm that can learn from data and to understand the limits of what can be learned from data.

→ One of the main goal of Computational learning theory is to develop algorithm that can learn from data. This includes both supervised and supervised learning algorithms.

→ Another goal of Computational learning theory is to understand the limits of what can be learned from data. This includes understanding of ~~data~~ Sample complexity of learning algorithm.

The Sample Comp. of learning algo. is the no. of data points that the algorithm needs to see in order

to learn a function. The sample complexity tells us how much data an algorithm need in order to learn a function.

→ goal of Comp. Lear. theory are to develop algorithm that can learn from data and to understand the limits of what can be learned from data.

Main methods of Computational learning theory.

- ① Inductive learning :- Most common type of learning in AI, a computer program is given a set of training examples and then tasked to generate a general rule that can be used to predict the O/P for new I/P.
- ② Deductive learning :- A comp program is given a set of rules or a model, and then tasked with using these to generate outputs for new I/Ps.
- ③ Abductive learning :- a comp. program is given a set of training data, and it is then tasked with generating a hypothesis (a possible explanation for the data) that is consistent with data.
- ④ Reinforcement learning :- given a set of reward and punishments and tasked is to maximize the reward and minimize punishments.

Challenges faced by Computational Learning Theory.

① One challenge is lack of data. In order to learn, AI needs data, However sometimes there is not enough data Available, or the data is not of good quality. This can make it difficult for AI to learn.

② Another challenge is that of Overfitting. This occurs when AI model learn too much from the training data, and as a result, does not generalise well to new data. This can lead to poor performance on tasks that the AI was not trained well.

③ Another challenge is that of Computational Complexity. Some learning tasks are simply too difficult for current AI technology to handle. This can make it difficult to develop effective AI models for the task.